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age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

yes = 9

no = 5

$$Info(D) = I(9,5) = -\frac{9}{14} \log_2 \frac{9}{14} - \frac{5}{14} \log_2 \frac{5}{14} = 0.940$$

age

age	buys_computer
<=30	no
<=30	no
31...40	yes
>40	yes
>40	yes
>40	no
31...40	yes
<=30	no
<=30	yes
>40	yes
<=30	yes
31...40	yes
31...40	yes
>40	no

<=30 31-40 >40

	yes	no
<=30	2	3
31-40	4	0
>40	3	2

$$Info_{age}(D) = \frac{5}{14} I(2,3) + \frac{4}{14} I(4,0) + \frac{5}{14} I(3,2) = 0.694$$

$$Gain(age) = Info(D) - Info_{age}(D)$$

$$Gain(age) = 0.940 - 0.694 = 0.246$$

income

income	buys_computer
high	no
high	no
high	yes
medium	yes
low	yes
low	no
low	yes
medium	no
low	yes
medium	yes
medium	yes
medium	yes
high	yes
medium	no

low med high

	yes	no
low	3	2
med	4	2
high	1	2

$$Info_{income}(D) = \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2) = 0.911$$

$$Gain(income) = Info(D) - Info_{income}(D)$$

$$Gain(income) = 0.940 - 0.911 = 0.029$$

student

student	buys_computer
no	no
no	no
no	yes
no	yes
yes	yes
yes	no
yes	yes
no	no
yes	yes
yes	yes
yes	yes
no	yes
yes	yes
no	no

yes no

	yes	no
yes	6	3
no	1	4

$$Info_{student}(D) = \frac{7}{14} I(6,1) + \frac{7}{14} I(3,4) = 0.7885$$

$$Gain(student) = Info(D) - Info_{student}(D)$$

$$Gain(student) = 0.940 - 0.7885 = 0.151$$

credit_rating

credit_rating	buys_computer
fair	no
excellent	no
fair	yes
fair	yes
fair	yes
excellent	no
excellent	yes
fair	no
fair	yes
fair	yes
excellent	yes
excellent	yes
fair	yes
excellent	no

fair excellent

	yes	no
fair	6	3
excellent	2	3

$$Info_{credit}(D) = \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3) = 0.992$$

$$Gain(credit) = Info(D) - Info_{credit}(D)$$

$$Gain(credit) = 0.940 - 0.992 = 0.048$$

အခြေခံ Rootnode

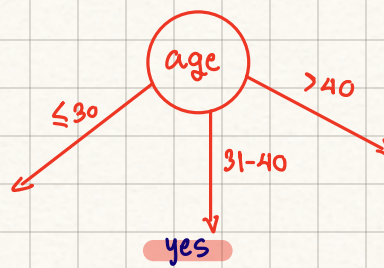
$$\text{Gain (age)} = 0.296$$

$$\text{Gain (income)} = 0.029$$

$$\text{Gain (student)} = 0.151$$

$$\text{Gain (credit)} = 0.048$$

age > student > Credit > income



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Student > Credit > income

age	student	buys computer
≤30	no	no
≤30	no	no
31...40	no	yes
>40	no	yes
>40	yes	yes
>40	yes	no
31...40	yes	yes
≤30	no	no
≤30	yes	yes
>40	yes	yes
≤30	yes	yes
31...40	no	yes
31...40	yes	yes
>40	no	no

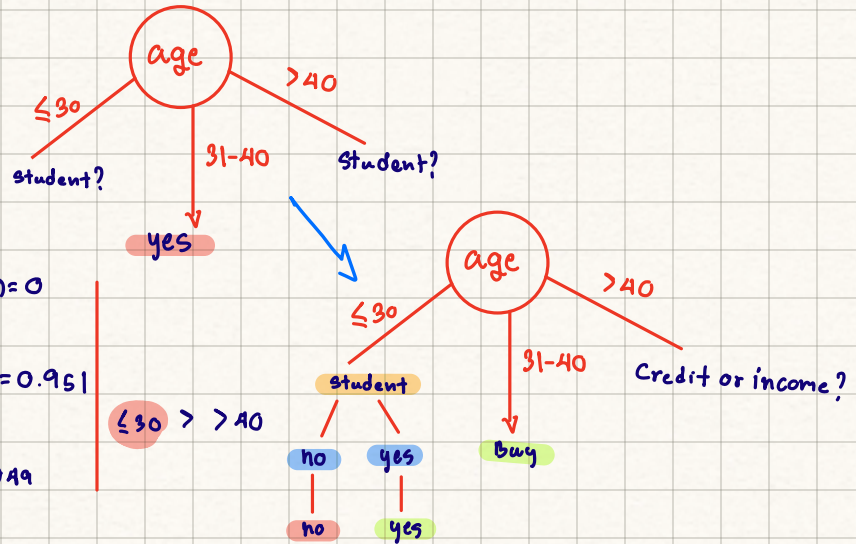
$$\text{Info}(D) = I(5,5) = 1$$

$$\text{Info}_{\leq 30}(D) = \frac{2}{5} I(2,0) + \frac{3}{5} I(0,3) = 0$$

$$\text{Info}_{>40}(D) = \frac{3}{5} I(2,1) + \frac{2}{5} I(1,1) = 0.951$$

$$\text{Gain}(\leq 30) = 1 - 0 = 1$$

$$\text{Gain}(>40) = 1 - 0.951 = 0.049$$



အခြေခံ Credit > income

age	income	student	credit rating	buys computer
≤30	high	no	fair	no
≤30	high	no	excellent	no
31...40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31...40	low	yes	excellent	yes
≤30	medium	no	fair	no
≤30	low	yes	fair	yes
>40	medium	yes	fair	yes
≤30	medium	yes	excellent	yes
31...40	medium	no	excellent	yes
31...40	high	yes	fair	yes
>40	medium	no	excellent	no

$$\text{Info}(D) = I(3,2) = -\frac{3}{5} \log_2 \frac{3}{5} - \frac{2}{5} \log_2 \frac{2}{5} = 0.970$$

$$\text{Info}_{\text{credit}}(D) = \frac{3}{5} I(3,0) + \frac{2}{5} I(0,2) = 0$$

$$\text{Info}_{\text{income}}(D) = \frac{3}{5} I(2,1) + \frac{2}{5} I(1,1) = 0.951$$

$$\text{Gain}(\text{credit}) = 0.970 - 0 = 0.970$$

$$\text{Gain}(\text{income}) = 0.970 - 0.951 = 0.019$$

Credit > income

